

Solving Problems Scientifically

A Comprehensive Technical Framework

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To TBD

*"An expert is a person who has made all the mistakes that can be made
in a very narrow field."* — Niels Bohr

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And $\text{\LaTeX}$ and $\text{\LaTeX} 2_{\epsilon}$.

Preface

Why did I write this book? TBD

Note to the Reader

Work in Progress

Please note that this book is currently a **work in progress**. Content, code snippets, and structural frameworks are actively being developed, refined, and updated. Chapters may be added, reorganized, or revised as the material matures. There is a companion [YouTube channel](#). I do not have a timeline for this work. It is a labour of love.

Document Conventions

Links are shown in:

- [blue](#) for links to other parts within this book, such as chapters, sections, glossary, and page numbers.
- [red](#) for links to the reference section.
- [orange](#) for links outside the book, such as a website.

Code and shell commands are shown in:

- Shell commands (zsh, bash) are shown inside a box outlined in black, with a note above it that identifies it as shell commands.
- Python commands are shown inside a box outlined in black, with a note above it that identifies it as code.

List of Figures

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Chapter 1

Introduction

Solving problems scientifically involves using the [scientific method](#), a structured, data-driven framework designed to eliminate guesswork and replace it with empirical evidence. [ADD MORE](#)

Chapter 2

Scientific method

The scientific method is doing something in a systematic and logical manner. A claim is verified through rigorous trails, not anecdotal evidence.

From one source:

- observation
- hypothesis
- experimentation
- conclusion

Other thoughts:

- what problem are you trying solve?
- establish methods of monitoring and control

What does Wikipedia say?

What does the Cambridge Dictionary say?

Appendix A

Python environment setup

A.1 Install Python for Arch Linux

On Arch Linux, Python is treated as a core system component. Do not install manual binaries from the website; instead, use the native [package manager](#), [pacman](#), to ensure rolling-release dependency tracking remains unified.

A.1.1 Procedural steps

1. Synchronize the package databases and upgrade the system infrastructure.
2. Explicitly install the core Python package.

A.1.2 Install Python with pacman

Identify as shell commands:

```
$ Shell: pacman
```

```
sudo pacman -Syu python
```

A.1.3 Verify the version

Validate the native binary deployment:

```
$ Shell: Python version
```

```
python --version
```

To exit the interactive environment, execute **exit()**.

A.1.4 Verify the environment

Regardless of your underlying host operating system, a successful deployment should allow entry into the interactive Python Read-Eval-Print Loop (REPL).

```
$ Shell: Run Python
```

```
python
```

Expected terminal output state:

\$ Shell: Python version

```
Python 3.14.6 (main, Jun 15 2026, 11:36:54) [GCC 16.1.1 20260430] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>
```

To exit the interactive environment, execute **exit()**.

A.2 Install Python for Windows

To establish a clean Python environment on Microsoft Windows, utilize the official installer provided by the Python Software Foundation.

A.2.1 Procedural steps

1. Navigate to the official download architecture at python.org.
2. Download the latest stable **Windows installer (64-bit)**.
3. Launch the executable installer.
4. **Critical Step:** At the bottom of the installation wizard window, check the box labeled **"Add python.exe to PATH"**. Failure to do this will prevent execution from standard terminals.
5. Select **"Install Now"**.
6. Once complete, verify the environmental path configuration by opening [PowerShell](#) or the Command Prompt and executing the verification command.

A.2.2 Verify the version

Identify as shell commands:

\$ Shell: Python version

```
python --version
```

To exit the interactive environment, execute **exit()**.

A.3 Install Python for macOS

While macOS includes a system stub for Python, development workflows require a dedicated, unencumbered framework binaries from the official source.

A.3.1 Procedural steps

1. Navigate to python.org.
2. Download the **macOS universal installer** package (.pkg).
3. Run the installer package and follow the standard system prompts.
4. The installer automatically handles path updates for standard shells.

A.3.2 Verify the version

Open your terminal (Zsh) and execute:

```
$ Shell: Python version
```

```
python3 --version
```


Appendix B

Appendix—dawgdad

B.1 Install dawgdad

dawgdad (Pilon 2026) is a **Python** package for statistical, graphical, and predictive analysis.

dawgdad requires Python. Install a **Python** distribution appropriate to your operating system (**Python environment setup**).

Install dawgdad following the instructions at <https://github.com/gillespilon/dawgdad>. dawgdad is compatible with the **latest Python distribution** from **Python.org**. Instructions on how to use dawgdad are provided in relevant sections of this document. Full documentation is available at <https://dawgdad.readthedocs.io/en/latest/>.

Run the following command in your shell:

\$ Shell: Installing dawgdad

```
pip install dawgdad
```

If you prefer a different distribution, such as **Anaconda**, following the instruction as appropriate.

B.2 Using dawgdad in Python

Here is a quick example of how to import and initialize the module:

▷ **Code: Initializing the data framework**

```
1 import dawgdad as dd
2
3 # Load and analyze your dataset
4 data = dd.load_analytics("performance_log.csv")
5 print(data.summary())
```


Glossary

A

Arch Linux Arch Linux is a [open-source software](#), rolling release Linux distribution. ([Arch Linux](#))

C

computer program A computer program is a sequence of instructions written in a [programming language](#) that is executed by a computer.

D

datum A datum is an individual value in a collection of data, discrete or continuous values that convey information, describing the quantity, quality, fact, [statistic](#), other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. Data are usually organized into structures such as tables that provide additional context and meaning, and may themselves be used as data in larger structures.

E

empirical method It is any procedure for conducting an investigation that relies upon experimentation and systematic [observation](#) rather than theoretical speculation.

experiment It is a series of [observations](#) conducted under controlled conditions to study a relationship with the purpose of drawing causal inferences about that relationship. An experiment involves the manipulation of an independent variable, the measurement of a dependent variable, and the exposure of various participants to one or more of the conditions being studied. Random selection of participants and their random assignment to conditions also are necessary in experiments.

O

observation It is the careful, close examination of an object, process, or other phenomenon for the purpose of collecting data about it or drawing conclusions.

open-source software It is computer software that is released under a license in which the copyright holder grants users the rights to use, study, change, and distribute the software and its source code to anyone and for any purpose. Open-source software may be developed in a collaborative public manner. Open-source software is a prominent example of open collaboration, meaning any capable user is able to participate online in development, making the number of possible contributors indefinite. The ability to examine the code facilitates public trust in the software.

P

package manager A package manager is software that handles package installation, removal, and downgrades of software for the host system. ([Package manager](#)) [5](#)

pacman Pacman is an [Arch Linux package manager](#) that handles package installation, removal, downgrades, and automatic dependency resolution. ([Pacman](#)) [5](#)

PowerShell PowerShell is a task automation and configuration management framework from Microsoft, consisting of a command-line shell and a scripting language. [6](#)

programming language A programming language is a system of notation for writing [computer programs](#).

Python Python is an easy to learn, powerful [programming language](#). It has efficient high-level [data](#) structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. ([python](#)) [9](#)

S

scientific method It is an [empirical method](#) consisting of a set of procedures, guidelines, assumptions, and attitudes required for the organized and systematic collection, interpretation, and verification of data and the discovery of reproducible evidence, enabling laws and principles to be stated or modified. One researcher comes up with a result (from theory, hunch, [observation](#)). This is not the truth. A scientific, emergent truth is not the result of any one [experiment](#). Many researchers try to verify it. If no one can prove it wrong, one can declare a consensus of [observations](#) and [experiments](#) has emerged in the scientific community. ¹

statistic TBD

Endnotes

References

Pilon, Gilles (Jan. 24, 2026). *dawgdad*. Version 1.0.9. URL: <https://github.com/gillespilon/dawgdad>.

About the Author

Gilles Pilon, P.Eng., is a Senior Machine Learning Engineer and Data Scientist with an extensive background in engineering frameworks and advanced systems development. A graduate of McMaster University in Metallurgical Engineering, he specializes in high-performance computing architectures, custom automation pipelines, and robust software infrastructure.

When not structuring complex codebases or optimizing technical documentation frameworks, he is an avid cyclist focused on performance analytics and outdoor endurance. He lives with his family, continuing to build open-source tools and data utilities.

